

The Rice SIAM chapter 50-digit challenge*

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Abstract

The Rice University SIAM student chapter published five new 10-digit problems. The intention is that Rice students solve every problem to as many digits of precision as possible. Archetype for this idea was a similar competition which culminated in a beautiful book [1]. This note gives my solutions, an alternative sixth problem, and an appendix on the integral of $\sin^2(\tan(\tan(\pi x)))$, which is where the mathematics is.

Key words. high-accuracy computation, Markov chain, elliptic billiard, Rouché’s theorem, Taylor-series integration, elastic collision, contour shift, reproducible computation

AMS subject classifications. 00A08, 30E20, 60J10, 65D30, 65F15, 65L05

1. Introduction

Each of the questions below has an answer that is a single real number. The problems should be answered in the form $d_1.d_2 \dots d_{10} \times 10^p$, $d_1 \neq 0$, where the exponent must be correct but does not count toward the number of correct digits and d_{10} is obtained by rounding.

1. Two players each control one knight at opposite corners on an 8×8 chessboard. They simultaneously try to move their respective pieces to an open square, using the L-shaped jump of the knight. Each of the valid jumps for either of the knights has equal probability of occurring. However if the two players choose to move their knights to the same square, they instead leave their pieces where they are and the turn is passed. What is the probability of at least one knight being at one of the four corners of the chessboard after 2005 turns?
2. A photon lies in the x - y plane trapped between two elliptical mirrors given by the equations $4x^2 + y^2 = 4$ and $9x^2 + 16y^2 = 144$. At time $t = 0$ the photon begins heading due north from position $(2, 0)$ with unit speed. How far is the photon from the origin at time $t = 60$?

*Written in 2005; this is a corrected edition. Section 6 of the original gave the wrong answer, and Sections 3 and 5 drew the wrong conclusion from a right one. Section 1.1 lists everything that changed. The code is at `code/`, and `make assets` reproduces every number below.

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3. Define the polynomial $p(x) = \sum_{k=0}^{10,000} a_k x^k$, where a_k is the $(k+1)$ st prime number. What is the magnitude of the root of p nearest the origin?
4. A point mass is at rest at the origin in the x - y plane. At time $t = 0$ the four other point masses listed in the table below are traveling clockwise with unit speed perpendicular to the radial direction. Each mass m_i attracts each other mass m_j with a gravitational force of magnitude $F_{ij} = m_i m_j / r^2$, where r is the distance of m_i from m_j . If these are the only forces present, how far from the origin is point p_4 at time $t = 5$?

Point	p_0	p_1	p_2	p_3	p_4
m_i	10	1	1	1	1
x_i	0	0	2	0	-4
y_i	0	1	0	-3	0

5. A sphere with unit radius centered at the origin in an (x, y, z) coordinate system is missing the part of its surface with z -coordinate greater than $3/4$. Two uniform balls with radius $1/4$ and unit mass are at rest at positions $(1/2, 1/8, 0)$ and $(0, 1/2, 0)$ inside the open “fishbowl.” At time $t = 0$, the balls are imparted with unit velocities toward the origin. What is the distance between the centers of the two balls at $t = 10$ if all collisions are completely elastic and there are no external forces?

Each of the next sections covers one problem, and the answers are collected in Section 7. I could not resist adding a suggestion for a sixth problem, and in an appendix I discuss the integral

$$\int_0^1 \sin^2(\tan(\tan(\pi x))) dx,$$

which is a problem I had some fun with.

1.1. What changed in this edition

The original note is twenty years old and three of its sections needed work.

Section 6 was wrong. The two balls were made to exchange their whole velocity vectors on impact, which is the head-on case. The answer to Problem 5 is 0.5433103836, not the 1.016560194 submitted in 2005.

Section 3 drew the wrong moral. A table of Mathematica runs was read as showing that “a loss of 43 digits is unavoidable” in an elliptic billiard. What that table measured was the worst-case bound significance arithmetic carries along, not the conditioning of the problem, which costs about one and a half digits over the whole flight.

Section 5 certified its answer the wrong way. It perturbed the initial data and watched the spread. That measures conditioning; the error of the integration is shared by all those runs and cannot show up in their spread. Both experiments are now run, and separately.

Section 4 had a gap. The bound that licensed truncating the polynomial controls function values and says nothing about where the roots are. Rouché’s theorem fills it in two lines.

The MATLAB and Mathematica of 2005 has been rewritten in Python: Claude Opus 5 was asked to revisit the original article and redo its experiments. The listings in this note are extracted from the files that produce its numbers, so the two cannot drift apart, and every figure, table and displayed constant comes out of `make assets`. I have also fixed the typography and the typing, of which there was a good deal.

2. A Markov chain

We analyse the random walk of two knights on a chessboard.

Problem. *Two players each control one knight at opposite corners on an 8×8 chessboard. . . What is the probability of at least one knight being at one of the four corners of the chessboard after 2005 turns?* $\diamond$

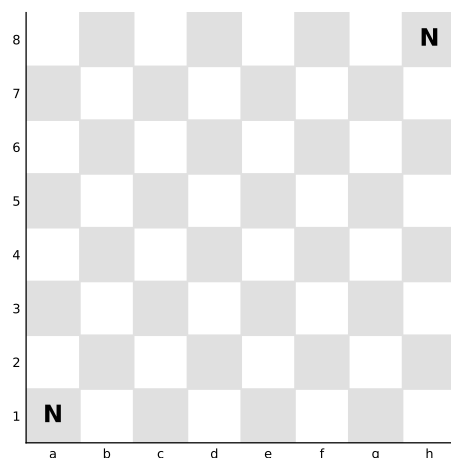


Figure 1: The knights start in opposite corners, on squares of the same colour.

To each square we assign a number $p = 8(i - 1) + j$, where i is the row and j the column, so the knights start on squares 1 and 64. A pair of occupied squares is a state, numbered $s = 64(p^{(1)} - 1) + p^{(2)}$, and there are $64^2 = 4096$ of them. Many are unreachable. Both knights move or, when they choose the same square, neither does, so the two squares keep the colour they started with and only the same-coloured pairs occur.

From a square p the knight can reach:

```
def moves(p):
    """The squares a knight standing on square p can jump to."""
    row, col = divmod(p - 1, 8)
    jumps = ((2, -1), (2, 1), (1, -2), (1, 2),
             (-2, -1), (-2, 1), (-1, -2), (-1, 2))
    return [8 * (row + dr) + (col + dc) + 1
            for dr, dc in jumps
            if 0 <= row + dr < 8 and 0 <= col + dc < 8]
```

Let the first knight have n_1 squares available and the second n_2 , where in each case the square held by the other knight is excluded. The players choose independently, so each of the $n = n_1 n_2$ pairs has probability $1/n$. If m of those pairs name the same square, the turn is passed, and the chain stays where it is with probability m/n . This defines a matrix A :

```
def transition():
    """The column-stochastic matrix of the chain on the 4096 states.

    Each player picks uniformly among the squares his knight can
    reach, the square held by the other knight excluded. If both
    pick the same square nobody moves and the turn is passed, which
    is the diagonal entry.
    """
    A = sp.lil_matrix((64 * 64, 64 * 64))
    for i in range(1, 65):
        for j in range(1, 65):
            if i == j or colour(i) != colour(j):
                continue # unreachable: they move together
            to_i = [q for q in moves(i) if q != j]
            to_j = [q for q in moves(j) if q != i]
            n = len(to_i) * len(to_j)
            passed = 0
            for a in to_i:
                for b in to_j:
                    if a == b:
                        passed += 1
                    else:
                        A[state(a, b) - 1, state(i, j) - 1] += 1 / n
            A[state(i, j) - 1, state(i, j) - 1] += passed / n
    return A.tocsc()
```

A is 4096×4096 with 53848 nonzero entries, all nonnegative, and the entries of a reachable column sum to one. Starting from the state $s = 64$ we want $v = A^{2005} e_{64}$, whose i th entry is the probability of being in state i after 2005 turns. We do not form A^{2005} :

```
def corner_indicator():
    """1 on every state with at least one knight on a corner."""
    w = np.zeros(64 * 64)
    for a in range(1, 65):
        for b in range(1, 65):
            if a in CORNERS or b in CORNERS:
                w[state(a, b) - 1] = 1.0
    return w
```

```

def solve():
    A = transition()
    v = np.zeros(64 * 64)
    v[state(*START) - 1] = 1.0
    for _ in range(TURNS):
        v = A @ v
    w = corner_indicator()
    return A, v, w @ v

```

Summing the entries of v gives 1 to fourteen places, which is the arithmetic checking itself. Adding the entries of v over the states with a knight on a corner gives

0.0484937545922106.

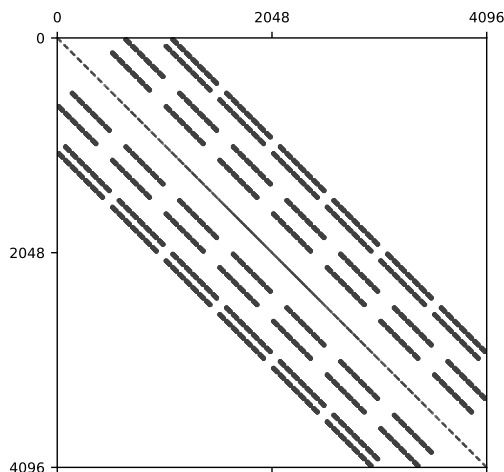


Figure 2: The 53848 nonzero entries of A .

The matrix has an eigenvalue exactly 1, and the rest lie in a disc of radius 0.9430904956 about the origin. Deleting the unreachable rows and columns leaves a column-stochastic matrix, whose dominant eigenvector u_1 is the stationary distribution; since $e_{64}^H u_1 \neq 0$ and $\|v\|_1 = 1$,

$$p = \frac{w^H u_1}{\|u_1\|_1},$$

with w the indicator of the corner states. This costs one eigenvector instead of 2005 matrix-vector products and agrees with the walk to twelve digits.

It also says something the walk does not. The subdominant eigenvalue has modulus 0.9430904956, and $0.9430904956^{2005} \approx 9.5 \cdot 10^{-52}$, so after 2005 turns nothing whatever is left of the initial condition: the answer is the stationary probability, and it would be the same after 300 turns or 300 000. The 2005 in the question is decoration. What does matter is the colour of the starting squares, since the chain never leaves the component it starts in.

Hence I submit as solution for this problem

0.04849375459.

3. A photon trap

We chase a photon trapped between two elliptic mirrors.

Problem. *A photon lies in the x - y plane trapped between two elliptical mirrors... How far is the photon from the origin at time $t = 60$?* $\diamond$

The inner mirror $4x^2 + y^2 = 4$ has semi-axes $a = 1$, $b = 2$ and the outer mirror $9x^2 + 16y^2 = 144$ has $a = 4$, $b = 3$. A moving photon was also an issue in the original competition, where Wagon discusses it in a chapter of [1].

Between reflections the photon travels in a straight line, so the time to the next mirror is the smallest positive root of a quadratic:

```
def hit(p, v, a, b, leaving):
    """Smallest  $t > 0$  with  $(px+t vx)^2/a^2 + (py+t vy)^2/b^2 = 1$ .

    'leaving' says the photon is sitting on this mirror, so  $t = 0$  is a
    root: divide it out rather than filter roots by a threshold.
    """
    qa = v[0] ** 2 / a ** 2 + v[1] ** 2 / b ** 2
    qb = 2 * (p[0] * v[0] / a ** 2 + p[1] * v[1] / b ** 2)
    if leaving:
        t = -qb / qa
        return t if t > 0 else mp.inf
    qc = p[0] ** 2 / a ** 2 + p[1] ** 2 / b ** 2 - 1
    disc = qb * qb - 4 * qa * qc
    if disc < 0:
        return mp.inf
    root = mp.sqrt(disc)
    ts = [t for t in ((-qb - root) / (2 * qa),
                     (-qb + root) / (2 * qa)) if t > 0]
    return min(ts) if ts else mp.inf
```

The one delicate point is the root at $t = 0$. After a reflection the photon sits on the mirror it just left, so that quadratic has a root there, and a routine returning the smallest positive root can return a rounding-error-sized one instead of the flight. Rather than reject roots below a threshold — which is a tolerance, and tolerances have to be tuned — we divide the known root out. The remaining root is $-b/a$ exactly.

At a reflection the tangential part of the velocity survives and the normal part changes sign. The outward normal of $x^2/a^2 + y^2/b^2 = 1$ at $\mathbf{p}$ is $(p_x/a^2, p_y/b^2)$, which is all we need:

```
def reflect(p, v, a, b):
    """Specular reflection at p on the ellipse with semi-axes a, b.

    The outward normal of  $x^2/a^2 + y^2/b^2 = 1$  at p is
    (px/a^2, py/b^2); the tangential part of v survives and the
    normal part changes sign.
    """
    n = (p[0] / a ** 2, p[1] / b ** 2)
    scale = mp.sqrt(n[0] ** 2 + n[1] ** 2)
    n = (n[0] / scale, n[1] / scale)
    s = 2 * (v[0] * n[0] + v[1] * n[1])
    return (v[0] - s * n[0], v[1] - s * n[1])
```

The flight is then a loop over events:

```
def fly(dps, shift=0, trace=False):
    """Distance from the origin at t = 60, at dps working digits."""
    mp.mp.dps = dps
    p = (mp.mpf(START[0]) + shift, mp.mpf(START[1]))
    v = (mp.mpf(DIRECTION[0]), mp.mpf(DIRECTION[1]))
    path, on, bounces, rem = [p], None, 0, mp.mpf(TEND)
    while rem > 0:
        dt, k = min((hit(p, v, a, b, on == i), i)
                    for i, (a, b) in enumerate(MIRRORS))
        dt = min(dt, rem)
        p, rem = (p[0] + dt * v[0], p[1] + dt * v[1]), rem - dt
        if trace:
            path.append(p)
        if rem > 0:
            v = reflect(p, v, *MIRRORS[k])
            on, bounces = k, bounces + 1
    return dict(distance=mp.sqrt(p[0] ** 2 + p[1] ** 2),
                bounces=bounces, path=path)
```

3.1. How ill-conditioned is it, really

Reflection off a curved mirror is the textbook amplifier of small differences, and the 2005 edition of this note ran the computation in Mathematica at a range of precisions, watched the number of certified digits fall far below the number carried, and concluded that “a loss of 43 digits is unavoidable.”¹

That is not what the experiment showed. Mathematica’s significance arithmetic carries an error bound along with each number and reports only the digits that bound can certify;

¹That conclusion was wrong, and the paragraph around it has been replaced. The submitted answer was not affected.

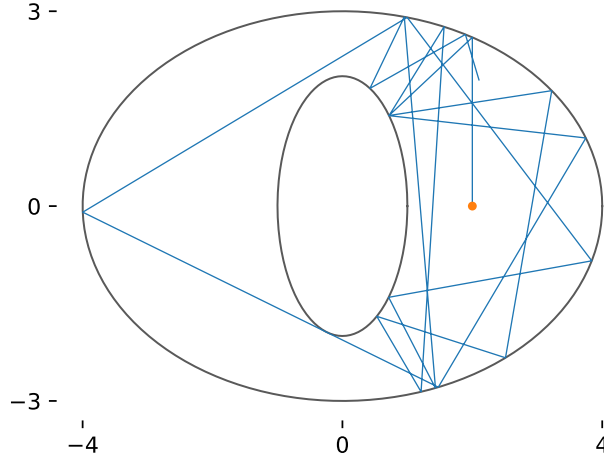


Figure 3: The path of the photon over $0 \leq t \leq 60$: 18 reflections, from the marked starting point.

the bound is worst-case and accumulates through every **Solve** and every **Inverse**. What the table measured was the growth of that bound, not the sensitivity of the answer.

Carrying a fixed number of digits instead, and comparing against a run at 140 digits, gives Table 3.1: the loss is a digit and a half, and it does not grow with the precision carried.

working digits	$ p(60) $	correct digits
20	2.8623358681628730953	18
25	2.8623358681628730962	23
30	2.8623358681628730962	28
40	2.8623358681628730962	38
50	2.8623358681628730962	49
70	2.8623358681628730962	68

Table 1: Correct digits in $|p(60)|$ against working precision.

The direct measurement agrees. Displacing the starting point by δ moves the answer by about 25.9δ , uniformly for δ from 10^{-40} down to 10^{-15} : over 18 reflections the amplification is a factor of 25.9, not 10^{43} . The billiard is mildly ill-conditioned, as promised, and that is all. Twelve working digits suffice for the ten the competition asks for; 2.86233586816287309621536776 is what 140 digits give, of which

2.862335868

is what I submit.

4. An eigenvalue problem

We compute the root nearest the origin of a polynomial of degree 10 000.

Problem. Define the polynomial $p(x) = \sum_{k=0}^{10,000} a_k x^k$, where a_k is the $(k + 1)$ st prime number. What is the magnitude of the root of p nearest the origin? $\diamond$

The roots of a polynomial are the eigenvalues of its companion matrix, and a degree-10 000 companion matrix is more than this problem deserves. The coefficients grow slowly — $a_{10000} = 104743$ — so for small $|x|$ the tail contributes very little, and we work with the truncation $p_M(x) = \sum_{k=0}^M a_k x^k$ at $M = 1993$:

```
def companion(coefficients):
    """Companion matrix of the monic multiple of the polynomial.

    coefficients[k] multiplies x^k, so the leading one is last.
    """
    a = np.array(coefficients, dtype=float)
    a = -a[:-1] / a[-1]
    n = len(a)
    A = np.zeros((n, n))
    A[0] = a[:-1]
    A[1:, :-1] = np.eye(n - 1)
    return A
```

On the circle $|x| = 0.82$ every omitted term is at most $a_{10000} 0.82^{M+1}$, and there are fewer than 10 000 of them:

```
def tail_bound(M=TRUNCATION, radius=CIRCLE):
    """A bound for  $|p - p_M|$  on  $|x| = \text{radius}$ .

    The coefficients increase, so every one of the omitted terms is at
    most  $a_{10000} \text{radius}^{(M+1)}$ .
    """
    mp.mp.dps = 40
    terms = DEGREE - M
    return terms * PRIMES[DEGREE] * radius ** (M + 1)
```

which gives $|p - p_M| \leq 1.16e - 17$ there.

4.1. Why the truncation is legitimate

The bound above controls the *values* of $p - p_M$. The question asks for a *root*, and a small perturbation of a function is not by itself a small perturbation of its zeros; the 2005 note passed over this.² What closes the gap is Rouché's theorem, which needs one more number: the minimum of $|p_M|$ on the same circle.

²A gap in the original argument, filled here. The answer is unchanged.

```

def minimum_modulus(M=TRUNCATION, radius=CIRCLE, samples=4096):
    """min |p_M| on the circle of the given radius.

    Sampled, then polished by a local search: the modulus of a
    polynomial has no flat minima on a circle, so a fine sample finds
    every basin.
    """
    mp.mp.dps = 40
    coefficients = [mp.mpf(q) for q in PRIMES[:M + 1]][::-1]
    def modulus(theta):
        z = radius * mp.expjpi(2 * theta)
        return abs(mp.polyval(coefficients, z))
    grid = [(modulus(mp.mpf(j) / samples), mp.mpf(j) / samples)
             for j in range(samples)]
    value, theta = min(grid)
    step = mp.mpf(1) / samples
    for _ in range(60):
        step /= 2
        for candidate in (theta - step, theta + step):
            trial = modulus(candidate)
            if trial < value:
                value, theta = trial, candidate
    return value

```

That minimum is 0.1268, larger than $1.16e-17$ by sixteen orders of magnitude, so $|p - p_M| < |p_M|$ on $|x| = 0.82$ and p and p_M have the same number of zeros inside. The truncation has 2, a conjugate pair, and therefore so does p ; the root nearest the origin is one of them, and no root of p hides in the disc unaccounted for.

An eigenvalue of the truncated companion matrix is now a starting point rather than an answer, and two Newton steps on the full polynomial settle it:

```

def polish(seed, dps=50):
    """Newton on the full degree-10000 polynomial, from a seed."""
    mp.mp.dps = dps
    a = [mp.mpf(q) for q in PRIMES][::-1]
    d = [a[i] * (DEGREE - i) for i in range(DEGREE)]
    return mp.findroot(lambda x: mp.polyval(a, x) / mp.polyval(d, x),
                       mp.mpc(seed), tol=mp.mpf(10) ** -40)

```

The root is a conjugate pair,

$$-0.6457580963469885 + 0.4831776762171610i,$$

of magnitude 0.8065135992606090061359, so I submit

$$0.8065135993.$$

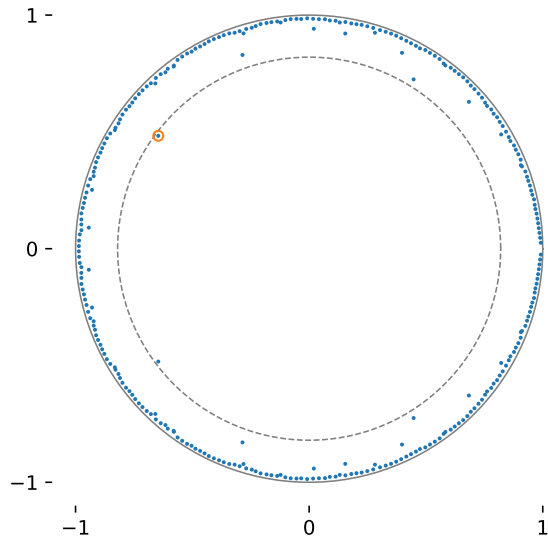


Figure 4: The spectrum of the companion matrix of p_{1993} , with the circles of radius 1 and 0.82. The nearest root is circled.

5. A five-body problem

The paths of five planets in a flat universe are determined by numerical integration.

Problem. *A point mass is at rest at the origin... how far from the origin is point p_4 at time $t = 5$?* $\diamond$

Identify the phase space with $\mathbb{C}^{10}$: let $u \in \mathbb{C}^5$ hold the positions and $v \in \mathbb{C}^5$ the velocities, so that

$$\frac{d}{dt} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ a \end{pmatrix}, \quad a_i = \sum_{j \neq i} m_j \frac{u_j - u_i}{|u_j - u_i|^3}, \quad (1)$$

the second equation being Newton's law with the masses divided out.

```
def field(u, v, masses=MASSES):
    """The acceleration of each mass, positions as complex numbers."""
    a = np.zeros(len(u), dtype=complex)
    for i in range(len(u)):
        for j in range(len(u)):
            if i != j:
                d = u[j] - u[i]
                a[i] += masses[j] * d / abs(d) ** 3
    return a
```

The initial velocities are clockwise, of unit length and perpendicular to the radius, so $u = (0, i, 2, -3i, -4)$ and $v = (0, 1, -i, -1, i)$. Equation (1) is not stiff and the flight is short; a good explicit method does it in a few seconds.

```

def integrate(rtol, shift=0.0, method="DOP853"):
    """|u_4(5)|, from a start optionally displaced by shift."""
    u0 = np.array(POSITIONS, dtype=complex)
    u0[WATCHED] += shift
    y0 = np.concatenate([u0.real, u0.imag,
                          np.real(VELOCITIES), np.imag(VELOCITIES)])

    def rhs(t, y):
        u = y[:5] + 1j * y[5:10]
        v = y[10:15] + 1j * y[15:]
        a = field(u, v)
        return np.concatenate([v.real, v.imag, a.real, a.imag])

    s = solve_ivp(rhs, [0, TEND], y0, method=method,
                  rtol=rtol, atol=1e-17)
    y = s.y[:, -1]
    return abs(complex(y[WATCHED], y[5 + WATCHED])), s.nfev

```

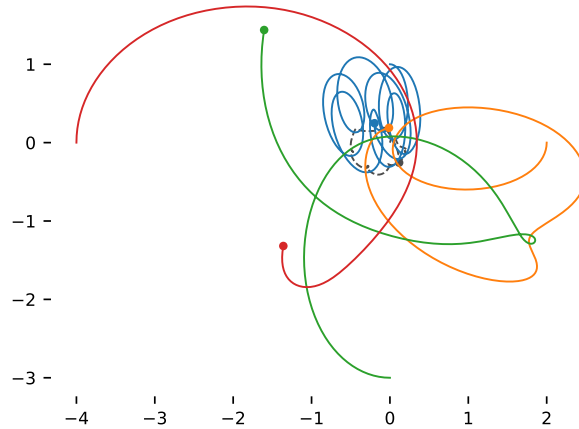


Figure 5: The five paths over $0 \leq t \leq 5$. The heavy mass, dashed, barely moves; the closest any pair comes is 0.0432.

5.1. How do we know ten digits are right

The 2005 note asked this question and answered it by perturbing the initial position of the fifth mass by about 10^{-16} , repeating the integration ten times, and reporting that the first

ten digits agreed every time.³

That experiment cannot answer the question. Every one of those runs used the same solver at the same tolerance, so they share their discretisation error almost exactly; what varies between them is the response to the data, and what is being asked about is the error of the method. A spread of 10^{-10} across the ten runs tells us the problem amplifies the data by about 26 over five time units — worth knowing, and not an answer to the question asked.

The experiment that does answer it varies the tolerance:

tolerance	$ u_4(5) $	evaluations
10^{-10}	1.89450969885495	10514
10^{-11}	1.89450978686529	13070
10^{-12}	1.89450979466496	15710
10^{-13}	1.89450979571379	19046
$3 \cdot 10^{-14}$	1.89450979576503	21698

Table 2: $|u_4(5)|$ against the tolerance asked of the integrator.

The digits settle from the left, as they should, and the last two rows agree to eleven. To be sure of the tenth we can leave floating-point arithmetic behind and integrate the Taylor series of the solution directly. The right-hand side of (1) is built from products and one power, so its Taylor coefficients satisfy recurrences: for $w = v^\alpha$,

$$n v_0 w_n = \sum_{k=0}^{n-1} (\alpha(n-k) - k) w_k v_{n-k},$$

which with $\alpha = -3/2$ is all that $|u_j - u_i|^{-3}$ requires.

```
def power_series(v, alpha, order):
    """Coefficients of v^alpha from the coefficients of v."""
    w = [v[0] ** alpha]
    for n in range(1, order + 1):
        acc = mp.mpf(0)
        for k in range(n):
            acc += (alpha * (n - k) - k) * w[k] * v[n - k]
        w.append(acc / (n * v[0]))
    return w
```

The coefficients of every coordinate follow, one order at a time:

```
def coefficients(x, y, vx, vy, masses, order):
    """Taylor coefficients of each coordinate about the state."""
    n_bodies = len(masses)
```

³The experiment was the wrong one; this subsection replaces it. The submitted answer was right, by a margin of one unit in the tenth digit.

```

X = [[xi] for xi in x]
Y = [[yi] for yi in y]
VX = [[vi] for vi in vx]
VY = [[vi] for vi in vy]
pairs = [(i, j) for i in range(n_bodies)
          for j in range(i + 1, n_bodies)]
R2 = {p: [] for p in pairs}
G = {p: [] for p in pairs}

for n in range(order + 1):
    for (i, j) in pairs:
        dx = [X[j][k] - X[i][k] for k in range(n + 1)]
        dy = [Y[j][k] - Y[i][k] for k in range(n + 1)]
        R2[(i, j)].append(
            sum(dx[k] * dx[n - k] + dy[k] * dy[n - k]
               for k in range(n + 1)))
        r2 = R2[(i, j)]
        if n == 0:
            G[(i, j)].append(r2[0] ** ALPHA)
        else:
            g = G[(i, j)]
            acc = sum((ALPHA * (n - k) - k) * g[k] * r2[n - k]
                      for k in range(n))
            g.append(acc / (n * r2[0]))
    if n == order:
        break
    ax = [mp.mpf(0)] * n_bodies
    ay = [mp.mpf(0)] * n_bodies
    for (i, j) in pairs:
        g = G[(i, j)]
        dx = [X[j][k] - X[i][k] for k in range(n + 1)]
        dy = [Y[j][k] - Y[i][k] for k in range(n + 1)]
        px = sum(dx[k] * g[n - k] for k in range(n + 1))
        py = sum(dy[k] * g[n - k] for k in range(n + 1))
        ax[i] += masses[j] * px
        ay[i] += masses[j] * py
        ax[j] -= masses[i] * px
        ay[j] -= masses[i] * py
    for i in range(n_bodies):
        X[i].append(VX[i][n] / (n + 1))
        Y[i].append(VY[i][n] / (n + 1))
        VX[i].append(ax[i] / (n + 1))
        VY[i].append(ay[i] / (n + 1))
return X, Y, VX, VY

```

The step is then chosen so that the last two terms of each series are at the level of the

tolerance, and the series is summed by Horner's rule.

```
def step_size(series, order, tol):
    """Jorba-Zou: match the last two terms of each series to tol."""
    h = mp.inf
    for s in series:
        for n in (order - 1, order):
            c = abs(s[n])
            if c > 0:
                h = min(h, (tol / c) ** (mp.mpf(1) / n))
    return h / 2
```

In 30-digit arithmetic at order 20 this gives

1.894509795783351650754,

and the energy is conserved to the same order, so the tenth digit is not in doubt. I submit

1.894509796.

6. Balls in the fishbowl

Two balls bounce around inside an open bowl.

Problem. *A sphere with unit radius centered at the origin... What is the distance between the centers of the two balls at $t = 10$?* $\diamond$

6.1. The problem is planar

Both centres start in the plane $z = 0$ with velocities in that plane, and every impulse the balls can receive lies in it. A ball touches the bowl where the ray from the origin through its centre meets the sphere, so the normal there is radial and the impulse is parallel to the centre's position vector; an impulse between the balls is parallel to the line joining their centres. The plane is invariant, and the missing cap $z > 3/4$ is never reached: it is a red herring.

What is left is two discs of radius $r = 1/4$. A centre touches the bowl at $|\mathbf{p}| = 1 - r = 3/4$ and the discs touch each other at $|\mathbf{p}_1 - \mathbf{p}_2| = 2r = 1/2$. Between contacts each centre moves in a straight line, so the event times are again roots of quadratics, and again the root at the surface just left is divided out rather than filtered:

```
def flight(dp, dv, target, leaving):
    """Smallest  $t > 0$  with  $|\mathbf{dp} + t \mathbf{dv}| = \text{target}$ , or infinity.

    'leaving' says the state is already on this surface, so  $t = 0$  is a
    root of the quadratic. Dividing it out leaves  $-b/a$  exactly and no
```

```

tolerance enters. After a collision it returns a negative number,
which correctly reports that the discs are separating.
"""
a = dot(dv, dv)
if a == 0:
    return mp.inf
b = 2 * dot(dp, dv)
if leaving:
    t = -b / a
    return t if t > 0 else mp.inf
c = dot(dp, dp) - target ** 2
disc = b * b - 4 * a * c
if disc < 0:
    return mp.inf
root = mp.sqrt(disc)
ts = [t for t in ((-b - root) / (2 * a), (-b + root) / (2 * a))
      if t > 0]
return min(ts) if ts else mp.inf

```

6.2. The collision

This is where the 2005 note went wrong.⁴ It argued in the centre-of-mass frame: with equal masses the two velocities there are $\pm \mathbf{w}/2$, where $\mathbf{w} = \mathbf{v}_1 - \mathbf{v}_2$, which is right; it then asserted that both “change their sign”, that is $\mathbf{w}^+ = -\mathbf{w}$, and concluded that the balls swap their velocity vectors.

Conservation of energy and momentum does not say that. It leaves the direction of $\mathbf{w}^+$ free and fixes only $|\mathbf{w}^+| = |\mathbf{w}|$. What chooses a direction is the geometry of the contact, and that ingredient was missing.

Let the discs touch with centres $\mathbf{p}_1, \mathbf{p}_2$ and put

$$\mathbf{n} = \frac{\mathbf{p}_1 - \mathbf{p}_2}{|\mathbf{p}_1 - \mathbf{p}_2|}, \quad \mathbf{w} = \mathbf{v}_1 - \mathbf{v}_2.$$

The surfaces are smooth, so the contact transmits no tangential force and the impulse is a multiple of $\mathbf{n}$. With unit masses, $\mathbf{v}_1^+ = \mathbf{v}_1 + \lambda \mathbf{n}$ and $\mathbf{v}_2^+ = \mathbf{v}_2 - \lambda \mathbf{n}$ conserve momentum for every λ , and elasticity,

$$|\mathbf{v}_1 + \lambda \mathbf{n}|^2 + |\mathbf{v}_2 - \lambda \mathbf{n}|^2 = |\mathbf{v}_1|^2 + |\mathbf{v}_2|^2,$$

reduces to $2\lambda(\mathbf{w} \cdot \mathbf{n}) + 2\lambda^2 = 0$. The root $\lambda = 0$ is the absence of a collision, and the other gives

$$\mathbf{v}_1^+ = \mathbf{v}_1 - (\mathbf{w} \cdot \mathbf{n})\mathbf{n}, \quad \mathbf{v}_2^+ = \mathbf{v}_2 + (\mathbf{w} \cdot \mathbf{n})\mathbf{n}, \quad (2)$$

that is $\mathbf{w}^+ = \mathbf{w} - 2(\mathbf{w} \cdot \mathbf{n})\mathbf{n}$: the relative velocity is reflected in the tangent plane at the contact, exactly as a velocity is reflected at the wall. The normal component reverses and the tangential component is untouched.

⁴The original derivation and its answer, 1.016560194, have been replaced. This is the one place where a submitted number changes.

```

def collide(p1, p2, v1, v2):
    """Elastic impact of two smooth unit masses in contact.

    The impulse is a multiple of the line of centres  $\mathbf{n}$ , and
    conservation of energy fixes the multiple. The normal part of
    the relative velocity reverses, the tangential part does not.
    """
    n = unit((p1[0] - p2[0], p1[1] - p2[1]))
    s = dot((v1[0] - v2[0], v1[1] - v2[1]), n)
    return ((v1[0] - s * n[0], v1[1] - s * n[1]),
            (v2[0] + s * n[0], v2[1] + s * n[1]))

```

Reversal, $\mathbf{w}^+ = -\mathbf{w}$, agrees with this if and only if $\mathbf{w}$ is parallel to $\mathbf{n}$, the head-on collision; otherwise it demands an impulse with a tangential component, which a frictionless contact cannot supply. Since reversal does conserve energy and momentum, neither of the checks the note made could see the difference.

Two consequences are worth naming. First, (2) does not preserve the individual speeds, only the sum of their squares, so the 2005 code's habit of rescaling both velocities to unit length after every event was consistent only with the rule it used. Second, the first contact here is 1.39 degrees from head-on, so the two rules very nearly agree at that one event; after 10 contacts and 18 rebounds nothing is left of the resemblance.

```

def run(dps=40, rule=collide, shift=0, trace=False):
    """Distance between the two centres at  $t = 10$ ."""
    mp.mp.dps = dps
    wall = 1 - RADIUS # a centre touches the bowl here
    touch = 2 * RADIUS # the centres touch here
    p1 = (mp.mpf(1) / 2 + shift, mp.mpf(1) / 8)
    p2 = (mp.mpf(0), mp.mpf(1) / 2)
    v1, v2 = unit((-p1[0], -p1[1])), unit((-p2[0], -p2[1]))

    path1, path2 = [p1], [p2]
    on, walls, contacts, angle, rem = None, 0, 0, None, mp.mpf(TEND)
    while rem > 0:
        dp = (p1[0] - p2[0], p1[1] - p2[1])
        dv = (v1[0] - v2[0], v1[1] - v2[1])
        dt, kind = min((flight(p1, v1, wall, on == "w1"), "w1"),
                       (flight(p2, v2, wall, on == "w2"), "w2"),
                       (flight(dp, dv, touch, on == "c"), "c"))
        if dt >= rem:
            dt, kind = rem, "stop"
        p1 = (p1[0] + dt * v1[0], p1[1] + dt * v1[1])
        p2 = (p2[0] + dt * v2[0], p2[1] + dt * v2[1])
        rem -= dt
        if trace:

```

```

        path1.append(p1)
        path2.append(p2)
    if kind == "stop":
        break
    if kind == "w1":
        v1, walls = reflect(p1, v1), walls + 1
    elif kind == "w2":
        v2, walls = reflect(p2, v2), walls + 1
    else:
        if angle is None:
            angle = impact_angle(p1, p2, v1, v2)
        v1, v2 = rule(p1, p2, v1, v2)
        contacts += 1
    on = kind
return dict(distance=norm((p1[0] - p2[0], p1[1] - p2[1])),
            walls=walls, contacts=contacts, angle=angle,
            energy=(dot(v1, v1) + dot(v2, v2)) / 2,
            path1=path1, path2=path2)

```

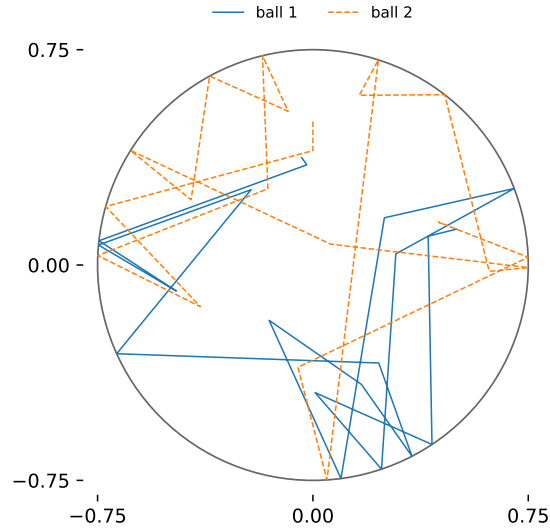


Figure 6: The paths of the two centres up to $t = 10$, inside the circle $|\mathbf{p}| = 3/4$ to which they are confined.

Displacing a ball by δ moves the answer by about 2191δ , so three to four digits are lost and fourteen working digits are enough. At 25, 40 and 60 digits the computation returns 0.5433103836074288213617 each time, the kinetic energy is still 1 at the end, and I submit

0.5433103836.

7. Conclusions

The problems provided some fun, although Problems 2 and 5 are too closely related: both are billiards, and the code for the first was recycled for the second. That recycling is how the error in Section 6 got in. The rebound off a curved wall carried over intact, and the one genuinely new ingredient — what happens when the two balls meet — was the piece that never got a derivation. An alternative problem is stated in the appendix.

Problem	Result
1	0.04849375459
2	2.862335868
3	0.8065135993
4	1.894509796
5	0.5433103836

Table 3: The five answers. Problem 5 is not the number submitted in 2005.

The numbers come with a guarantee this time: each is reproduced by `make assets`, at two precisions or by two methods.

References

- [1] F. Bornemann, D. Laurie, S. Wagon and J. Waldvogel, *The SIAM 100-Digit Challenge: A Study in High-Accuracy Numerical Computing*, SIAM, Philadelphia, 2004.
- [2] F. Bornemann and T. Schmelzer, *Taming a hydra of singularities*, Amer. Math. Monthly **114** (2007), no. 8, 727–732.
- [3] L. N. Trefethen and D. Bau, III, *Numerical Linear Algebra*, SIAM, Philadelphia, 1997.

Appendix A. A sixth problem

There might be a tie at the end of the competition, and an additional problem would then be published. Here is my suggestion.

Problem. *Compute*

$$\int_0^1 \sin^2(\tan(\tan(\pi x))) \, dx. \tag{3}$$

It was Brian Davies who pointed out that the evaluation would involve serious difficulties both numerically and analytically. It is hard to integrate this function on the real line, but a small step into the complex plane gives us everything. The argument sketched here was afterwards worked out with Folkmar Bornemann and published in the Monthly [2], where the integrand is one member of a family and the closed form below is one value of a limit.

A.1. Existence

The function $f(x) = \sin^2(\tan(\tan(\pi x)))$ is undefined at some points of $[0, 1]$: the tangent has poles at $\pi/2 + k\pi$, so f is undefined on the countable set

$$X = \left\{ \frac{1}{\pi} \arctan\left(\frac{\pi}{2} + k\pi\right), k \in \mathbb{Z} \right\} \cup \left\{ \frac{1}{2} \right\}.$$

Since $0 \leq f \leq 1$ on $[0, 1] \setminus X$, the function is integrable and the integral lies between 0 and 1. That is the last easy statement about it on the real line: the graph oscillates without bound between consecutive points of X , and a quadrature rule sees a sequence of essentially random samples. Averaging over an equispaced grid of 100000 points returns 0.391037, which has one correct digit.

A.2. The integrand in the complex plane

From $e^{iz} = \cos z + i \sin z$,

$$\sin^2 z = \left(\frac{e^{iz} - e^{-iz}}{2i} \right)^2 = \frac{1}{2} - \frac{1}{4}(e^{2iz} + e^{-2iz}) = \frac{1}{2} - \frac{1}{2} \cos(2z),$$

and for real x this gives the two real forms

$$\sin^2 x = \frac{1}{2} - \frac{1}{2} \Re e^{2ix} = \frac{1}{2} - \frac{1}{2} \Re e^{-2ix},$$

which agree because $\Re \bar{z} = \Re z$. The functions e^{2iz} and e^{-2iz} inside them continue to the whole plane; $\sin^2 z$ is not their real part off the real line, and we shall not need it to be. The first is bounded in the upper half-plane, where $|e^{2iz}| = e^{-2\Im z} \leq 1$, and the second in the lower half-plane. Write

$$f_c^+(x) = e^{+2i \tan(\tan(\pi x + ic))}, \quad f_c^-(x) = e^{-2i \tan(\tan(\pi x - ic))}.$$

Lemma A.1 *The complex tangent maps the open upper half-plane into itself and the open lower half-plane into itself.*

Proof. Let $z = x + iy$. Then $\tan z = \sin z \overline{\cos z} / |\cos z|^2$, and from

$$\sin z = \sin x \cosh y + i \cos x \sinh y, \quad \cos z = \cos x \cosh y - i \sin x \sinh y$$

the numerator has imaginary part $\sinh y \cosh y (\sin^2 x + \cos^2 x) = \sinh y \cosh y$. Hence

$$\Im \tan z = \frac{\sinh y \cosh y}{|\cos z|^2},$$

which has the sign of y . The poles of the tangent are real, so $\tan$ is analytic on either open half-plane. $\square$

By Lemma A.1 applied twice, $\tan(\tan z)$ is analytic and takes values in the upper half-plane when z does, so f_0^+ is analytic and bounded by 1 there. The same holds for f_0^- below the axis.

A.3. Dominated convergence

For $c > 0$ the function f_c^+ is continuous, and as $c \rightarrow 0$ it converges to $e^{2i \tan(\tan(\pi x))}$ at every $x \in [0, 1] \setminus X$, hence almost everywhere. It is bounded by 1 throughout, so dominated convergence gives

$$\int_0^1 f_c^+ dx \xrightarrow{c \rightarrow 0} \int_0^1 e^{2i \tan(\tan(\pi x))} dx, \quad (4)$$

and the same argument applies to f_c^- below the axis.

A.4. Cauchy's theorem

The integrand f_0^+ is analytic in the open upper half-plane, so its integral along a closed path there vanishes. Take the rectangle with corners ic_1 , $1 + ic_1$, $1 + ic_2$, ic_2 for $0 < c_1 < c_2$, with sides A , C vertical and B , D horizontal.

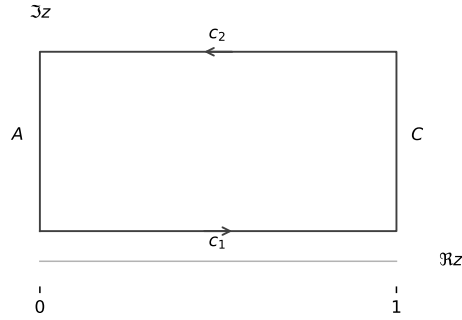


Figure 7: The path of integration.

Since $\tan(\pi(x+1)+ic) = \tan(\pi x+ic)$, the integrand has period one and the two vertical sides cancel. What is left is

$$\int_0^1 f_{c_1}^+ dx = \int_0^1 f_{c_2}^+ dx,$$

so the integral does not depend on the height of the contour:

$$\int_0^1 f_c^+ dx = C \quad \text{for all } c > 0,$$

and by (4) the same constant is the integral along the real axis.

The constant is now read off at the other end. As $c \rightarrow \infty$, $\tan(\pi x + ic) \rightarrow i$ uniformly in x , so $\tan(\tan(\pi x + ic)) \rightarrow \tan i = i \tanh 1$ uniformly and

$$f_c^+(x) \longrightarrow e^{2i \cdot i \tanh 1} = e^{-2 \tanh 1},$$

again uniformly. The integrands are bounded by 1, so the integrals converge to the same limit, and

$$C = e^{-2 \tanh 1}.$$

Keeping in mind how badly f_0^+ behaves on the real line, it is a pleasant surprise that lifting the contour turns it into the simplest function there is.

```
def closed_form(dps=40):
    """(1 - e^{-2 tanh 1})/2, the value the contour argument gives."""
    mp.mp.dps = dps
    return (1 - mp.e ** (-2 * mp.tanh(1))) / 2
```

Putting the pieces together,

$$\int_0^1 \sin^2(\tan(\tan(\pi x))) dx = \frac{1}{2} - \frac{1}{2}e^{-2 \tanh 1} = 0.390992162151530454532\dots,$$

and the same argument applied to f_c^- in the lower half-plane gives the same number, as it must.

A.5. A numerical confirmation

The argument makes two claims that can be checked separately: that $\int_0^1 f_c^+$ does not depend on c , and that its value is $e^{-2 \tanh 1}$. For $c > 0$ the integrand is analytic and periodic, so the trapezoid rule converges geometrically and a few dozen points suffice.

```
def lifted(c, n, dps=40):
    """The trapezoid rule for the integral of f_c^+ over one period.

    f_c^+(x) = exp(2i tan(tan(pi (x + i c)))) is analytic and has
    period 1 for c > 0, so the trapezoid rule converges
    geometrically and the answer must not depend on c at all.
    """
    mp.mp.dps = dps
    total = mp.mpc(0)
    for j in range(n):
        z = mp.mpf(j) / n + mp.mpc(0, 1) * c
        total += mp.e ** (2j * mp.tan(mp.tan(mp.pi * z)))
    return total / n
```

At $c = 1$ and $c = 1/4$, sixty-four points already return 0.3909921622..., and at $c = 1/20$, where the integrand is beginning to oscillate, two hundred and fifty-six points do. The contour can be lowered until the trapezoid rule fails, and the answer does not move until it does.

To ten digits, the sixth problem has the answer

$$0.3909921622.$$

The same contour shift handles $\sin^2$ of any number of nested tangents, and the values approach a limit that [2] identifies; the problem above is the second member of that family.